

WHOLE GENOME SEQUENCING

WGS Pipeline

Candida albicans

Antifungal Resistance Variant Analysis

SRR5876982

Reference SC5314

Diploid (2n)

38.17× coverage

Background & Objectives

Candida albicans is an opportunistic diploid fungus behind most systemic candidiasis, especially in immunocompromised patients. Resistance to azoles (fluconazole) and echinocandins (caspofungin) now threatens therapy.

WGS enables comprehensive detection of resistance-linked variants - across drug targets (**ERG11**, **FKS1**), efflux regulators (**TAC1**), and genomic events such as LOH and aneuploidy.

OBJECTIVES

- 01 Run the WGS pipeline end-to-end, from raw FASTQ reads to an annotated VCF
- 02 Identify variants (SNP, indel, MNP) with FreeBayes in diploid mode
- 03 Functionally annotate variants with a custom snpEff database
- 04 Profile resistance genes: ERG11, FKS1, TAC1, CDR1, CDR2
- 05 Document the technical challenges encountered and how they were solved

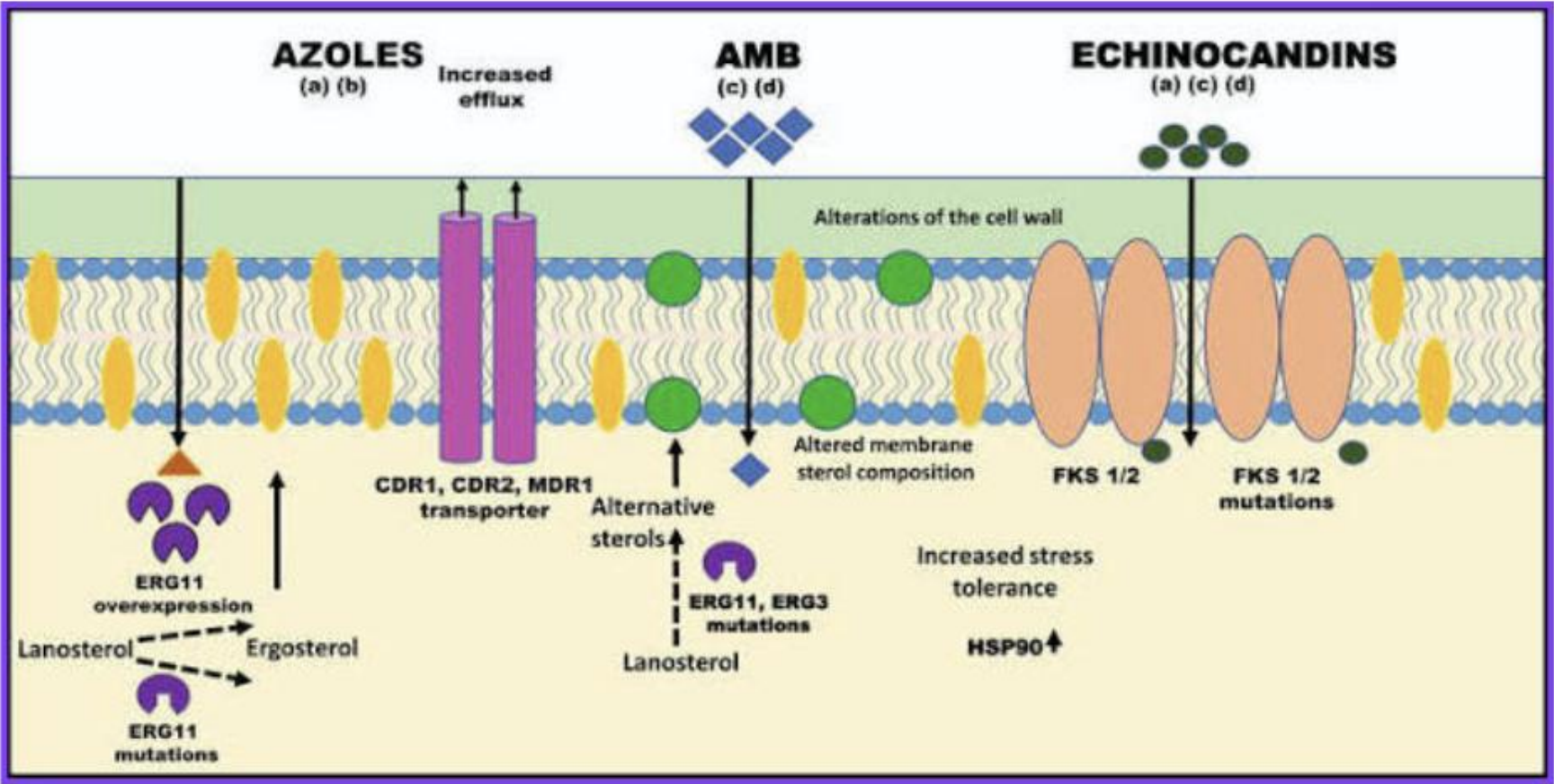
Diploidy & Resistance Mechanisms

C. albicans is obligately diploid — each gene has two alleles, so VCF genotypes carry biological meaning:

GT	Interpretation
0 / 0	Homozygous ref — both alleles wild-type
0 / 1	Heterozygous — one mutant allele · early resistance
1 / 1	Homozygous alt — both mutant · Loss of Heterozygosity

RESISTANCE MECHANISMS

- **ERG11** mutations (lanosterol 14α-demethylase) — direct fluconazole target
- **CDR1/CDR2** efflux-pump overexpression driven by TAC1 gain-of-function
- **FKS1** hotspot HS1 (S645) / HS2 — direct echinocandin target
- **Loss of Heterozygosity** — heterozygous converts to homozygous mutant
- **Chromosomal aneuploidy** — duplication of resistance-bearing chromosomes



(Ruiz-Baca et al., 2020)

Why ERG11 & FKS1 Are Drug Targets

ERG11 Azole · fluconazole

- Encodes **lanosterol 14 α -demethylase** — key enzyme of ergosterol biosynthesis, a fungal membrane component
- Azoles inhibit it → ergosterol fails to form → membrane breaks down
- Active-site mutations (Y132H, K143R, F145L) → drug can't bind, enzyme still works → **resistant**

FKS1 Echinocandin · caspofungin

- Encodes the catalytic subunit of **β -1,3-glucan synthase** — builds the fungal cell wall
- Echinocandins inhibit it → weakened cell wall → cell lysis
- Mutations cluster at hotspots HS1 (641–649) & HS2 (1345–1365) , the drug-contact domain → **resistant**

WHY SELECTIVE?

Both gene products are **essential for fungal survival** , and both act on fungal-specific pathways absent in human cells — making them selective, safe drug targets. (Ruiz-Baca et al., 2021)

A Fluconazole-Passaged Isolate

DATASET

Accession	SRR5876982 · ENA
Sample	C. albicans FL22-passaged
Reference	SC5314 GCF_000182965.3
Genome	8 chromosomes · 14.3 Mb
Platform	Illumina HiSeq 2500 Paired-end, 100 bp
Ploidy	DIPLOID (2n)

WHY FL22-PASSAGED?

Fluconazole-passaged **22 times** — each passage a sub-inhibitory exposure cycle. The accumulated selection pressure makes this isolate ideal for studying the molecular evolution of antifungal resistance.

TOOLCHAIN · CONDA ENV CANDIDA

FastQC 0.12	MultiQC 1.34	fastp 1.3	BWA-MEM 0.7	SAMtools
FreeBayes	bcftools	snpEff 5.4c		

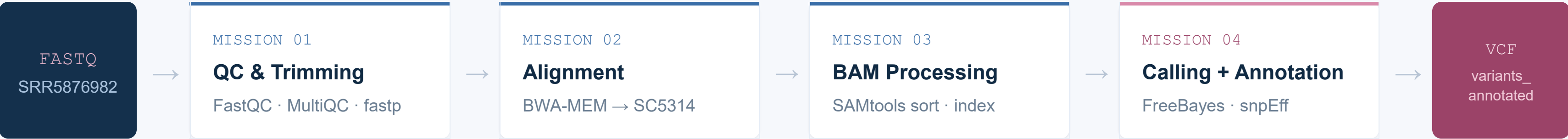
01

METHODS

The Four-Mission Pipeline

WORKFLOW

From Raw Reads to Annotated VCF



6.6M

raw reads input (R1 + R2)

94.9%

reads mapped to reference

38.17×

mean coverage depth

63,875

annotated variants

Clean Reads with fastp

```
// adapter trimming + quality filtering

$ fastp -i sample_1.fastq -I sample_2.fastq \
  -o clean_1.fastq -O clean_2.fastq \
  --detect_adapter_for_pe \
  --qualified_quality_phred 20 \
  --unqualified_percent_limit 40 \
  --n_base_limit 5 --length_required 50 \
  --thread 4 --html report.html
```

KEY PARAMETERS

phred 20	Min Q20 (1% error)
unqual 40%	Tolerated low-Q bases / read
n_base 5	Drop reads with >5 N bases
length 50	Min length after trimming

MISSION 02 - 03 · Alignment & BAM Processing

Map to SC5314, Sort & Index

```
// align with read group, then BAM pipeline

$ bwa mem -t 4 -R '@RG\tID:..\tSM:Ca_FL22\tPL:ILLUMINA' \
  reference.fna clean_1.fastq clean_2.fastq > aln.sam

$ samtools sort -@ 4 aln.bam -o sorted.bam

$ samtools index sorted.bam

$ samtools flagstat sorted.bam
```

WHY THE READ GROUP MATTERS

FreeBayes requires **@RG** tags for accurate variant calling. The tag records **ID** (run), **SM** (sample), **PL** (platform) and **LB** (library) — without it, downstream calling fails or mislabels reads.

Diploid Calling, Then Annotate

```
// Bayesian calling - DIPLOID

$ freebayes -f reference.fna \
  --min-coverage 5 --ploidy 2 \
  sorted.bam > raw.vcf

$ bcftools filter -e 'QUAL<20 || DP<5' raw.vcf

$ snpEff -Xmx6g ann CaSC5314_custom \
  norm.vcf > annotated.vcf
```

CRITICAL · PLOIDY

C. albicans is **diploid**. Using `--ploidy 1` would break genotype interpretation and lose the heterozygosity essential to resistance analysis.

CUSTOM SNPEFF DATABASE

Built directly from `reference.fna` + the matching NCBI GFF3 — public databases use a fragmented assembly incompatible with our 8-chromosome reference.

02

RESULTS

Quality, Coverage & Variants

Clean, High-Quality Reads

6 . 6M

raw reads (R1 + R2)

88%

reads passed filter

5 . 4%

duplication rate

170bp

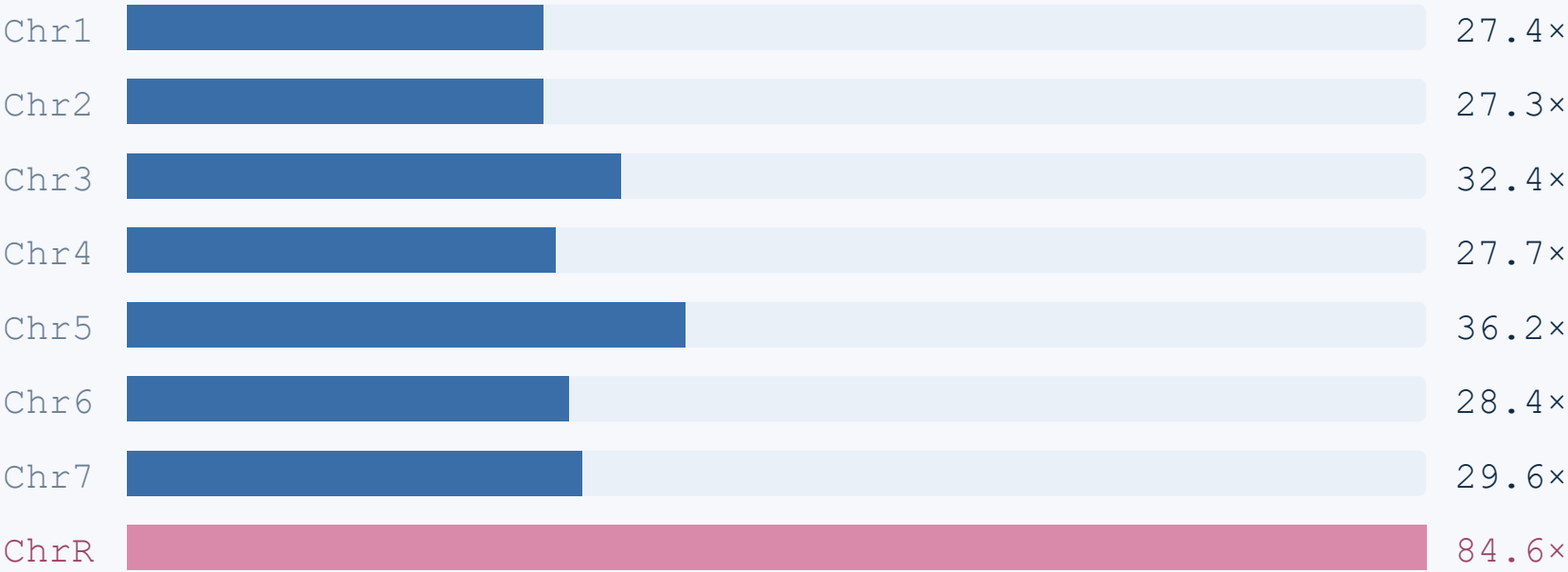
insert size peak

METRIC	BEFORE	AFTER
Q20 (R1 / R2)	90 . 6 / 94 . 5	95 . 9 / 97 . 4
Q30 (R1 / R2)	85 . 1 / 91 . 0	92 . 6 / 95 . 1

Quality of the FL22-passaged data is markedly better than clinical datasets — it comes from a pure laboratory culture with no host-DNA contamination, and fastp detected no adapter .

94.9% Mapped, 38× Mean Depth

MEAN DEPTH PER CHROMOSOME



All chromosomes ≥ 99.9% covered. ChrR's high depth is collapsed rDNA repeat — an artifact, not real coverage.

Jones et al. (2004)

95.3%

reads mapped

38.17×

mean depth

75.6%

properly paired

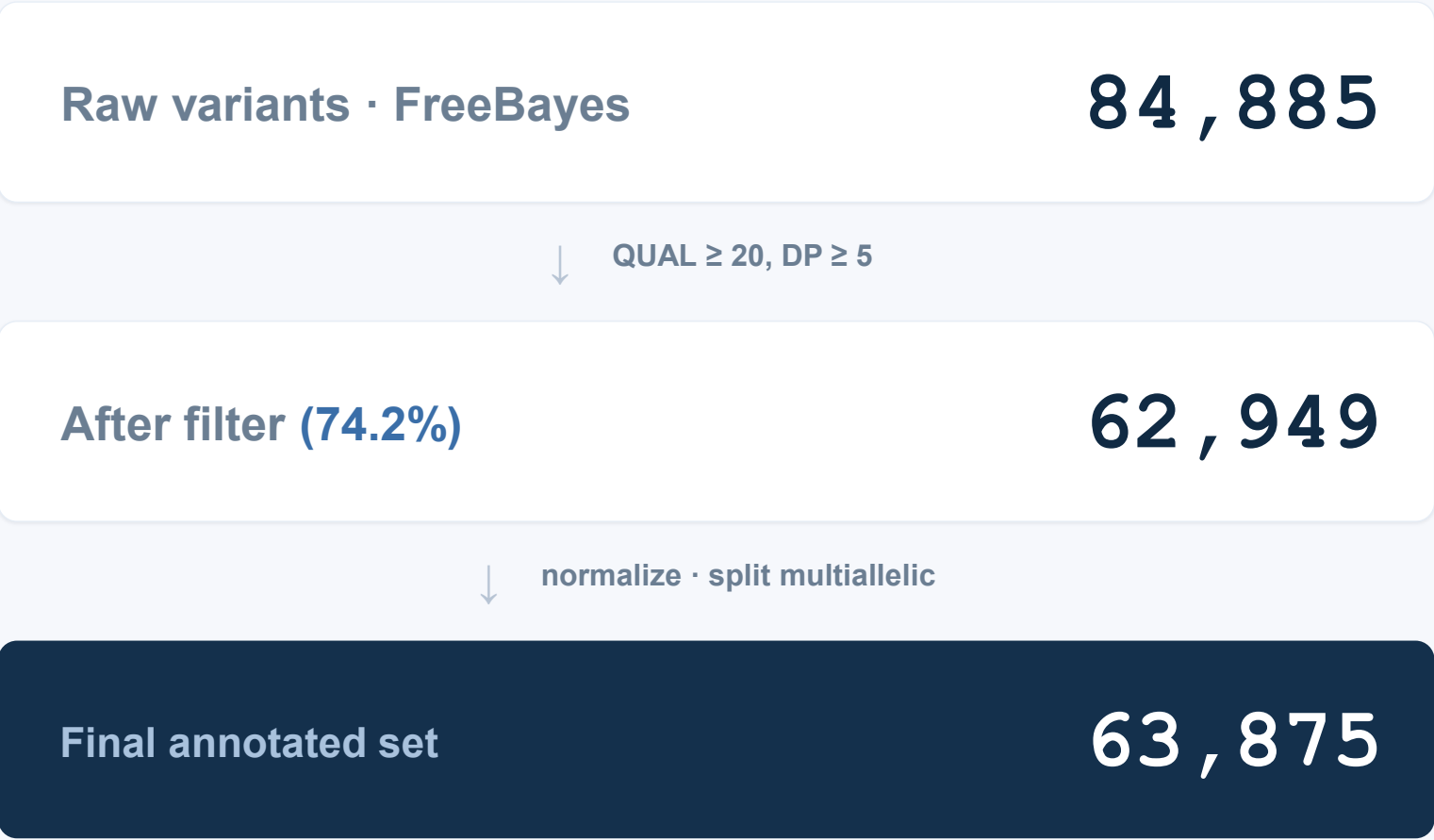
0.44%

singletons

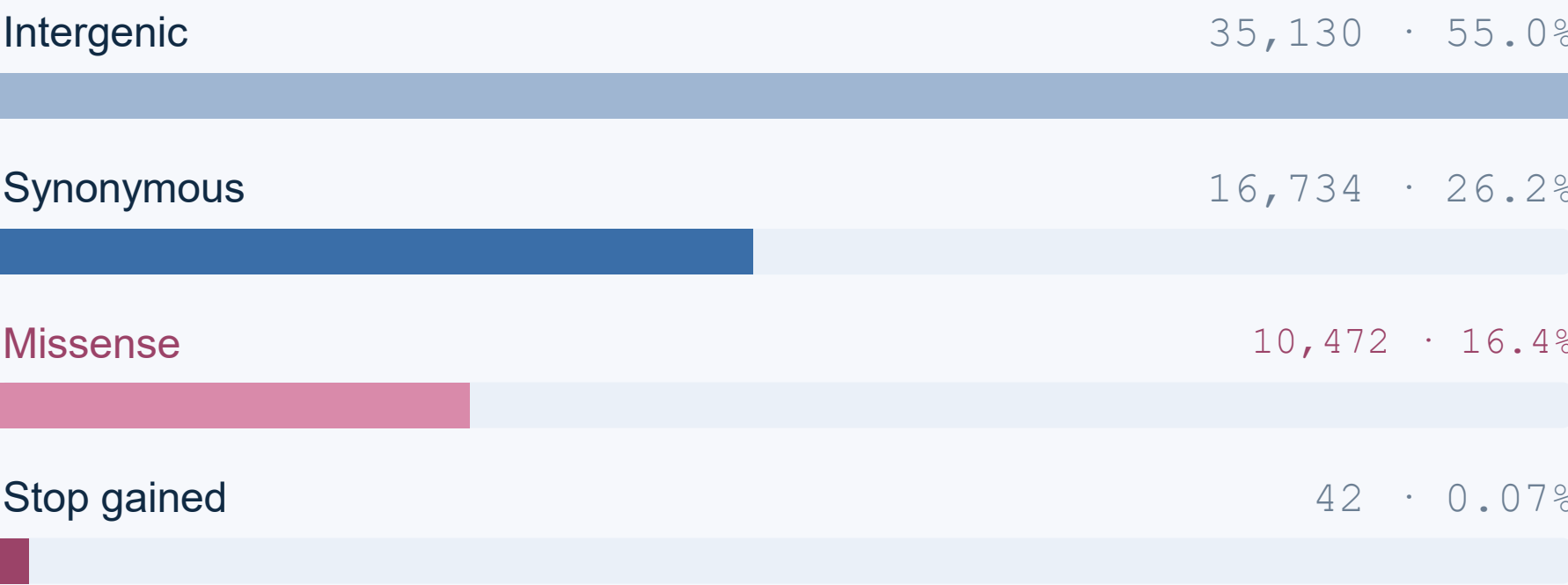
Mapping rate **94.9%** vs. just **16.5%** for a clinical *Plasmodium* set — pure culture means no host contamination.

63,875 Annotated Variants

CALLING & FILTERING FUNNEL



FUNCTIONAL EFFECT DISTRIBUTION



03

RESISTANCE PROFILE

Which Gene Carries Resistance?

Findings Gene by Gene

Gene	Drug Target	Status	Variants	Significance
GSC1 / FKS1	Echinocandin (caspofungin)	WILD-TYPE	13 (2 mis)	No variants in hotspot HS1 or HS2
ERG11	Azole (fluconazole)	WILD-TYPE	10 (2 mis)	K128T & D116E validated non-resistant in literature
TAC1	Azole (CDR1/2 regulator)	INCONCLUSIVE	23 (9 mis)	9 missense incl. p.Leu941Pro — needs RT-qPCR
CDR1 / CDR2	Azole (efflux pump)	NOT ANALYZED	—	Requires GFF coordinates + RT-qPCR

FKS1 & ERG11 — Wild-Type Confirmed

GSC1 / FKS1

WILD-TYPE

Echinocandin target · 1,897 aa · Chr1

Hotspot HS1	(641–649)	no variants
Hotspot HS2	(1345–1365)	no variants
Missense found		T1886S · P1838A

Both missense lie **far outside** the hotspots — consistent with an azole-only exposure history, no echinocandin pressure.

ERG11

WILD-TYPE

Fluconazole target · Chr5 · lanosterol 14α-demethylase

Variants found	10	(2 missense)
Missense		K128T · D116E

Both substitutions are characterized as non-resistant — Flowers et al. (2015) tested this exact combination in SC5314 and classed the allele wild-type.

Hassan 2013 · Flowers 2015

TAC1 — Significance Inconclusive

23 VARIANTS · 9 MISSENSE

CHANGE	QUAL	GT
p.Asp776Asn	446.6	0/1
p.Asn396Ser	438.8	0/1
p.Ser935Leu	232.2	0/1
p.Leu941Pro	222.2	0/1
+ 5 more missense, 14 synonymous		

Coordinates verified: NC_032093.1:416,400-419,345 (Chr5)

p.Leu941Pro introduces a proline — a rigid residue that may disrupt protein conformation. But a single variant can't be called gain-of-function from sequence alone.

Unlike FKS1 and ERG11, TAC1 has **no widely-validated hotspot catalog** — its variants are judged by measuring CDR1/CDR2 expression, not by position. (Mahdizade *et al.*, 2024)

Next step: RT-qPCR of CDR1 / CDR2 to test whether these variants drive efflux-pump overexpression.

Technical Challenges & Solutions

CHALLENGE	SOLUTION
snpEff OutOfMemoryError on 63,875 variants	Allocate larger Java heap — <code>-Xmx6g</code>
Public snpEff DB uses a fragmented assembly	Build custom DB from reference.fna + matching GFF3
Custom DB build failed (no .bin produced)	Build with <code>-noCheckCds</code> <code>-noCheckProtein</code>
ChrR coverage abnormally high (84.6×)	rDNA repeat collapse — use median of Chr1–7

CONCLUSIONS

What the Pipeline Tells Us

- 01 Reproducible 4-mission pipeline, raw FASTQ → annotated VCF, at **38.17× mean coverage**
- 02 `--ploidy 2` is **mandatory** for diploid *C. albicans*
- 03 A **custom snpEff database** was required for annotation against the modern reference
- 04 Both drug targets — **ERG11** and **FKS1** — are functionally wild-type

CENTRAL FINDING

With both drug targets wild-type, any resistance in this isolate most likely runs through **efflux-pump overexpression (CDR1/CDR2 via TAC1)** — not target-site mutation.

RECOMMENDATIONS

Next Steps

PRIORITY

RT-qPCR of CDR1 / CDR2 / TAC1

Test whether TAC1 missense variants drive efflux overexpression

PRIORITY

Susceptibility testing (EUCAST / E-test)

Confirm phenotype for fluconazole & caspofungin

FOLLOW-UP

LOH & copy-number analysis

YMAP / bedtools — resistance without point mutation

FOLLOW-UP

Compare to isogenic FL0 isolate

Identify de-novo mutations from fluconazole passage

+ Structural modeling of TAC1 p.Leu941Pro

+ Characterize GSC1 variants T1886S & P1838A

References

- Flowers SA, Colón B, Whaley SG, Schuler MA, Rogers PD (2015).** Contribution of clinically derived mutations in ERG11 to azole resistance in *Candida albicans*. *Antimicrobial Agents and Chemotherapy*, 59(1), 450–460. DOI: 10.1128/AAC.03470-14
- Jones T, Federspiel NA, Chibana H, Dungan J, Kalman S, Magee BB, Newport G, Thorstenson YR, Agabian N, Magee PT, Davis RW, Scherer S (2004).** The diploid genome sequence of *Candida albicans*. *Proceedings of the National Academy of Sciences*, 101(19), 7329–7334. DOI: 10.1073/pnas.0401648101
- Mahdizade Ari AH, Hoseinnejad A, Ghazanfari M, Boozhmehrani MJ, Bahreiny SS, Abastabar M, Galbo R, Giuffrè L, Haghani I, Romeo O (2024).** The TAC1 Gene in *Candida albicans*: Structure, Function, and Role in Azole Resistance: A Mini-Review. *Microbial Drug Resistance*, 30(7), 288–296. DOI: 10.1089/mdr.2023.0334
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- Ruiz-Baca E, Arredondo-Sánchez RI, Corral-Pérez K, López-Rodríguez A, Meneses-Morales I, Ayala-García VM, Martínez-Rocha AL (2021).** Molecular Mechanisms of Resistance to Antifungals in *Candida albicans*. Dalam: *Candida and Candidiasis*. IntechOpen. DOI: 10.5772/intechopen.95108